

11. Spectral emission lines are unique to each element. These lines are produced when electrons make transitions between two energy levels and photons of light are emitted as a result.

The diagram represents some of the energy levels for a hydrogen atom.

$$E_3 \text{ ————— } -1.36 \times 10^{-19} \text{ J}$$

$$E_2 \text{ ————— } -2.42 \times 10^{-19} \text{ J}$$

$$E_1 \text{ ————— } -5.45 \times 10^{-19} \text{ J}$$

$$E_0 \text{ ————— } -21.8 \times 10^{-19} \text{ J}$$

(a) For the energy levels shown:

- (i) determine the number of possible emission lines caused by the transition of electrons between the energy levels

1

- (ii) identify the electron transition that results in the emission of a photon with the shortest wavelength.

1



* X 8 5 7 7 6 0 1 3 6 *

11. (continued)

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(b) An electron makes the transition from E_3 to E_1 .

(i) Determine the wavelength of the photon of light emitted.

4

Space for working and answer

(ii) State the colour of this photon of light.

1

(c) The image below shows part of the line emission spectrum for a hydrogen atom. The red emission line is brighter than the others.



Explain why the red emission line is brighter than the others.

2

[Turn over



* X 8 5 7 7 6 0 1 3 7 *